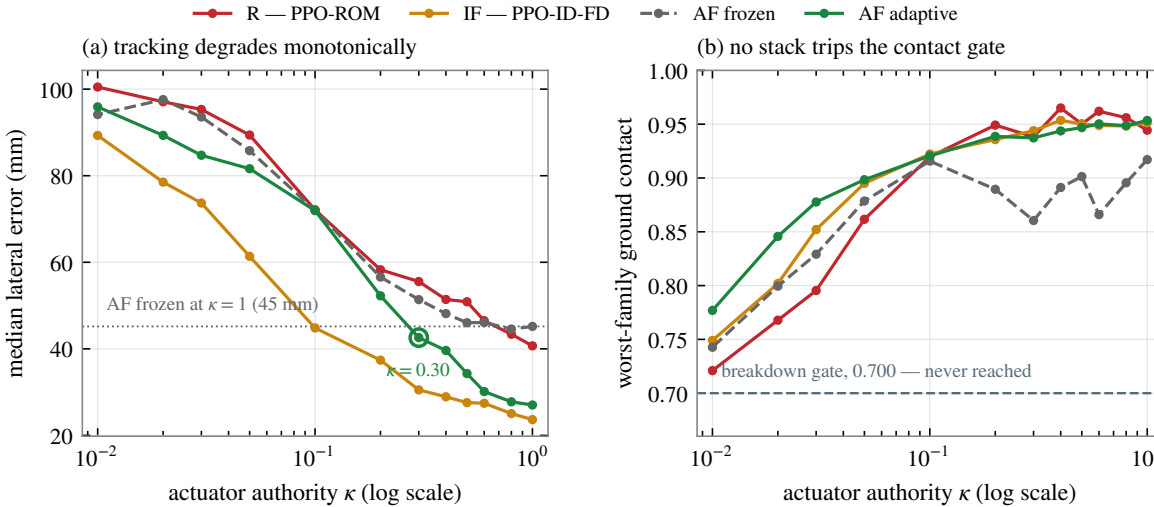
R — PPO-ROM
IF — PPO-ID-FD
AF frozen
AF adaptive
(a) tracking degrades monotonically
median lateral error (mm)
AF frozen at κ = 1 (45 mm)
κ = 0.30
actuator authority κ (log scale)
(b) no stack trips the contact gate
worst-family ground contact
breakdown gate, 0.700 — never reached
actuator authority κ (log scale)